C-DP.1.3 · Weekly check · 5 questions

MA.7.DP.1.3 · Predictions from a random sample · Unit 5 Scaling With Proportional Relationships · Given Thursday, on the week's benchmark

Form title C-DP.1.3 Check · Predictions from a random sample
Description Learning goal: I can use a random sample to make a prediction about a population. 5 questions, 1 point each. Work each one out on paper, then choose or type your answer. Calculator allowed.
Locked mode OFF (calculator allowed; Locked mode blocks the Chromebook calculator)

Short-answer validation (already on the template's short-answer question; copy that question): Regular expression → Matches → `^-?(\d+(\.\d+)?|\.\d+|\d+/\d+|\d+ \d+/\d+)$`
Error text: Type only a number, like 12 or -3.5 or 2/7 or 4 1/3. No units, $, %, π or commas.

#	Question image	File · type	Options to add	Answer key	Pts
1	In a random sample of 100 passengers, 35 check a bag. Based on this sample, about how many of the 800 passengers in the population check a bag? A. 2,286 B. 35 C. 280 D. 520	**Q1.png** Multiple choice Same problem, now multiple choice (paper: multiple choice)	A B C D	**C**	1
2	To predict how many of the 4000 residents in the town support building a new park, Priya asked the 24 people at the skate park on Saturday, and 22 said yes. She predicts that about 3670 of the 4000 residents support building a new park. Why is this prediction not valid? A. A prediction about the town can only be made by asking every one of the 4000 residents B. The prediction should be 22 residents, the number who said yes, not 3670 residents C. The sample is too small; asking a larger group of people at the skate park on Saturday would make the prediction valid D. The people at the skate park on Saturday are not a random sample of all the residents, so they may not represent the whole town	**Q2.png** Multiple choice	A B C D	**D**	1
3	A library has 1200 readers. Based on a random sample of 240 readers, it is predicted that about 480 of the 1200 readers borrow e-books. How many readers in the sample said they borrow e-books? Type a number only. No units, $ or %.	**Q3.png** Short answer	none (typed answer)	**96** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
4	A factory checks a random sample of 400 marbles and finds that 100 of them are red. Based on this sample, about how many of 1000 marbles would be expected to be red? Type a number only. No units, $ or %.	**Q4.png** Short answer	none (typed answer)	**250** Tick "Mark all other answers incorrect." Response validation: Regular expression, Matches (see top).	1
5	The principal wants to predict how many of the 2100 students in the school would buy a school T-shirt. Two random samples of students were surveyed. Sample A: 600 students, and 210 said yes. Sample B: 120 students, and 54 said yes. The two samples give different predictions. Which sample's prediction should the principal trust more, about how many of the 2100 students does it predict, and why? A. Sample B, which predicts about 945 students, because a larger share of its students said yes. B. Sample A, which predicts about 735 students, because a larger random sample usually gives a proportion closer to the whole school's. C. Sample A, which predicts about 1365 students, because a larger random sample usually gives a proportion closer to the whole school's. D. Neither; the principal should use about 840 students, halfway between the two predictions, because the samples disagree.	**Q5.png** Rigor Multiple choice Same problem, now multiple choice (paper: a written answer)	A B C D	**B**	1
6	LEARNING GOAL I can use a random sample to make a prediction about a population. 4 I can judge whether a prediction is reasonable and explain how sample size or bias affects it. 3 I can use a proportion from a sample to predict a value for the whole population. 2 I can tell whether a sample is random and representative. 1 I need help to start this.	**Level scale.png** Multiple choice (the template's last question)	Level 1 Level 2 Level 3 Level 4	no key	0

Answers with worked solutions are in the Answers folder next to the question images (same file names).